

ATRIK SAMANTA

Reg No.: 23BAI0136 ◊ Vellore Institute of Technology (VIT), Vellore ◊ [linkedin.com/in/atrik-samanta-49a0603b6](https://www.linkedin.com/in/atrik-samanta-49a0603b6)
+91-7001913750 ◊ samanta.atrik1710@gmail.com ◊ github.com/ATRIK171005

EDUCATION

B.Tech in Computer Science and Engineering (AI & ML) Vellore Institute of Technology (VIT), Vellore	2023 - 2027 CGPA: 7.9
Class XII (CBSE) – Hem Sheela Model School	2023
Class X (CBSE) – Hem Sheela Model School	2021

SKILLS AND INTERESTS

Languages	Python, R Language, SQL, Java, C/C++, JavaScript, HTML/CSS
AI Tools & Agents	Claude Code, Model Context Protocol (MCP), GitHub Copilot, Cursor, LLMs (Llama, Qwen)
Optimization & ML Frameworks & Web	Google OR-Tools (Linear Programming, VRPTW), PyTorch, TensorFlow, Scikit-learn FastAPI, React, Next.js, Node.js, Streamlit, Vite, LangChain
Cloud, DevOps & DB	AWS (EC2), Docker, Git, Linux, Nginx, PostgreSQL, SQLite, Supabase, ChromaDB
Data & Analytics	NumPy, pandas, OpenCV, Matplotlib, Data Quality Assessment, Lineage Tracking
Core CS	Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming
Soft Skills	Stakeholder Communication, Process Automation, Cross-Functional Teamwork, Problem Solving
Spoken Languages	English, Hindi, Bengali, German (Level 1)

PROJECTS

Enterprise AI Copilot <i>Python, FastAPI, LangChain, OpenAI API, ChromaDB, PostgreSQL, Streamlit, Docker</i>	July 2026 – Present
- Developed an enterprise-grade AI Copilot enabling employees to interact with organizational knowledge through natural language using Retrieval-Augmented Generation (RAG). - Built a document intelligence pipeline by processing PDFs, generating vector embeddings, and storing them in ChromaDB for semantic search and context retrieval. - Implemented a FastAPI backend with role-based access control, conversation memory, and secure REST APIs alongside an interactive Streamlit interface for Q&A. - Optimized retrieval accuracy and response latency through document chunking, embedding generation, and vector similarity search, creating a scalable enterprise knowledge assistant.	
HoloSecEdge: Secure Edge Framework for Holographic Consumer IoT <i>Python, Machine Learning, Elliptic Curve Cryptography (ECC), Random Forest, Edge Computing</i>	July 2026 – Present
- Developed a lightweight edge-cloud security framework for holographic IoT devices, including AR/VR headsets, smart displays, and motion-sensing wearables. - Implemented Security-enriched Elliptic Curve Cryptography (SECC) to provide low-overhead encryption, authentication, and data integrity for resource-constrained edge devices. - Built an Adaptive Squirrel Search-driven Regularized Random Forest (ASS-RRF) model to detect anomalies and cyberattacks in real time. - Designed a trust-gated retraining mechanism to mitigate data poisoning attacks and evaluated performance on a public multi-source holographic IoT dataset across detection accuracy, latency, and energy metrics.	

Vehicle Routing Problem with Time Windows (VRPTW)
Python, Google OR-Tools, SQLite, Streamlit, Folium, Plotly, OSRM

July 2026 – August 2026

- Engineered an automated VRPTW optimization engine using Google OR-Tools to solve complex multi-constraint logistics workflows under vehicle capacity, customer time-windows, and route duration bounds.
- Prototyped interactive Streamlit analytics dashboards using Claude Code and Python for scenario testing, route mapping, and Plotly Gantt schedule tracking to measure delivery KPIs.
- Curated relational SQLite data schemas and integrated OSRM network matrices with disjunction-based infeasibility handling to eliminate routing bottlenecks across transport networks.

AI-Based Logistics Network Optimizer
Python, Google OR-Tools, SQLite, Streamlit, Plotly, NetworkX, Machine Learning

June 2026 – July 2026

- Developed a supply chain network optimization system using Linear Programming (OR-Tools) to automate transportation allocation and warehouse capacity management.
- Defined structured SQLite backend schemas to enforce data quality thresholds, track shipment data lineage, and maintain reliable data inputs across supply chain domains.
- Integrated ML-based demand forecasting models (Random Forest/XGBoost) into the optimization pipeline to enhance predictive planning accuracy and quantify operational productivity gains.

StructuralEye: Crack Detection System
Personal Project

March 2026 – April 2026

- Automated manual structural inspection workflows using computer vision algorithms including Canny edge detection, CLAHE, and morphological operations.
- Extracted quantitative physical dimensions (length, width, area) to compute dynamic risk scores (0–100) and establish baseline severity KPIs for defect monitoring.
- Visualized structural degradation trends over time via an interactive analytics dashboard to support data-driven maintenance decisions.

CERTIFICATIONS

IBM Advanced Generative AI Certification, IBM Career Education
Core Focus: Foundation Models, Transformers, LLMs, GANs, Diffusion Models, Advanced Prompt Engineering (Chain-of-Thought, Tree-of-Thoughts), and Agentic Deployment via IBM watsonx.ai, IBM Granite, Hugging Face, and GitHub Copilot.

May 2026 - July 2026

POSITIONS OF RESPONSIBILITY

Events Head
The Otaku Club

2024 - 2025

- Formulated, structured, and executed end-to-end interactive student events and operational workflows.
- Led cross-functional student teams across logistics, marketing, and operations to align stakeholder requirements and drive initiative execution.

ACHIEVEMENTS

- Built and deployed end-to-end AI/ML systems focused on Operations Research, Supply Chain Optimization, Computer Vision, and Generative AI/RAG Architectures.
- Elected and served as **Events Head** of The Otaku Club for the 2024–2025 academic year, managing event logistics and team delivery.